

João Barreiros Coelho Rodrigues

E-mail | Personal Webpage

 LinkedIn |  Github |  CiênciaVita |  ORCID |  Google Scholar

Lisbon, Portugal

RESEARCH INTERESTS

Digital Systems Design, Computer Architecture, Embedded Systems, Reconfigurable Computing, FPGA-based acceleration, High-Level Synthesis, and edge deployment of hardware-accelerated machine learning models.

EDUCATION

- **Instituto Superior Técnico – Universidade de Lisboa** 2026 – Present
PhD in Electrical and Computer Engineering Lisbon, Portugal
 - PhD Proposal: *Low-power, configurable Processing Unit for Embedded AI*
- **Instituto Superior Técnico – Universidade de Lisboa** Sep 2023 – Nov 2025
MSc in Electrical and Computer Engineering Lisbon, Portugal
 - GPA: **18/20**
 - Thesis: *Embedded System for Aerial Image Segmentation*, graded **19/20**
 - Major in Computer Systems; Minor in Electronic Circuits and Systems
- **Instituto Superior Técnico – Universidade de Lisboa** Sep 2020 – Feb 2024
BSc in Electrical and Computer Engineering Lisbon, Portugal
 - Final GPA: **14/20**

WORK EXPERIENCE

- **INESC-ID / INESC INOV** Nov 2025 – Present
Early-Stage Researcher Lisbon, Portugal
 - Conduct research on FPGA-based reconfigurable computing for efficient edge AI inference
 - Design and implement hardware accelerators using High-Level Synthesis and RTL
 - Assist on the development of methodologies for rapid HW/SW co-design targeting embedded systems
 - Contribute to the **SMARTChips** project, focusing on AI accelerator architectures and prototyping
 - Perform system-level validation through simulation and FPGA-based experimentation
- **Instituto Superior Técnico – Universidade de Lisboa** Nov 2023 – Aug 2025
Teaching Assistant Lisbon, Portugal
 - Taught laboratory and problem classes under the Department of Electrical and Computer Engineering
 - Courses lectured include Algorithms and Data Structures, Digital Systems, Programming, and Computer Architectures
 - Achieved consistently high student evaluations (up to 9/9 QUC scores)
 - Co-founded the **Open Lab for the Scientific Area of Computer Systems**, supporting hands-on undergraduate education

PUBLICATIONS

- Rodrigues, João Barreiros; Neto, Horácio C.; Véstias, Mário Pereira. "ARES: Dataflow Co-Design for Embedded Swin Transformer Acceleration on SoC-FPGAs." Accepted for publication in the Proceedings of the International Symposium on Highly Efficient Accelerators and Reconfigurable Technologies (HEART '26), Heidelberg, Germany, 2026. DOI: [10.1145/3814576.3814594](https://doi.org/10.1145/3814576.3814594)
- Teixeira, Rúben; Rodrigues, João Barreiros; Aguiar, Jaime; Li, Jiao; Sousa, José T. de. "Versat-AI: An ONNX-to-SoC Compiler for Model-Agnostic CGRA Edge Inference." *Journal of Low Power Electronics and Applications*, vol. 16, no. 3, article 36, 2026. DOI: [10.3390/jlpea16030036](https://doi.org/10.3390/jlpea16030036)

ACADEMIC PROJECTS

- **SMARTChips: FPGA-Based AI Accelerator for Embedded Systems**

Nov 2025 – Present

Tools: Brevitas, Py2HWSW, Vitis HLS, Vitis IDE, Vivado, Cadence

- Designing a reconfigurable neural processing unit (NPU) using High-Level Synthesis (HLS)
- Developing end-to-end C implementation of quantized ResNet8 for hardware/software co-validation

- **ARES: Accelerated Embedded Swin Transformer System**

Sep 2024 – Nov 2025

Tools: Brevitas, Vitis HLS, Vitis IDE, SoC-FPGA platform

- Developed hardware/software co-designed accelerator for Feed-Forward Network (FFN) of Swin Transformer
- Adapted vision transformer models for FPGA-friendly fixed-point inference
- Integrated accelerator with embedded software on SoC-FPGA for aerial image segmentation
- Achieved efficient resource utilization while maintaining inference accuracy
- Literature Review on various ViT projects for aerial object detection and segmentation

SKILLS

- **Hardware Description Languages:** C (HLS), VHDL, Verilog
- **Programming Languages:** C, C++, RISC-V Assembly, Python, MATLAB
- **Hardware Design & EDA:** Xilinx Vivado Suite, Cadence SoC Workflow
- **Operating Systems:** GNU/Linux
- **Developer Tools:** Git, Vim/Neovim, L^AT_EX
- **Other Hard Skills:** Linux Technical Support, Hardware Repair, Electronics Troubleshooting, Soldering

HONORS AND AWARDS

- **Academic Excellence**

2024/2025

Instituto Superior Técnico

- Awarded to A-level students ($\approx$ top 10%)

- **Teaching Excellence**

2023/2024, 2024/2025

Instituto Superior Técnico

- Awarded for the Algorithms and Data Structures, Digital Systems, Programming, and Computer Architectures courses

VOLUNTEER AND EXTRACURRICULAR EXPERIENCE

- **Co-manager**

Sep 2024 – Present

Lisbon, Portugal

DEEC OpenLab

- Founding member
- Workshop organizer
- Overall lab aid in electronics and 3D Printing fields

- **Board Member – Chief Hacker Officer**

Sep 2024 – Present

Lisbon, Portugal

HackerSchool

- Co-managed HackerSchool's technical infrastructure and laboratory
- Co-organized the **MakerIST** faire
- Organize technical and recruitment workshops
- Consulting on HackerSchool's decision making and projects

- **President**

Sep 2022 – Sep 2024

Lisbon, Portugal

HackerSchool

- Managed HackerSchool's financial and technical infrastructure
- Organized technical and recruitment workshops and promoted Free and Open Source Hardware/Software practices
- Coordinated partnerships with the **Electrical and Computer Engineering Department**
- Arranged sponsorship with the **Instituto Superior Técnico**
- Arranged participation on **Mini-MakerFaire Lisbon 2023**

- **Member**

Mar 2021 – Present

Lisbon, Portugal

HackerSchool

- Teaching workshops on Electronics, Embedded Systems, Git and Additive Manufacturing Processes
- Actively contribute to the development of various project prototypes

EDUCATION OUTREACH

- **Electronics First** May 2024 
MIT, DEEC-IST, HackerSchool
- **3D Modelling and Printing for ECE Majors** April 2025
DEEC Openlab, DEEC-IST, HackerSchool
- **Microcontrollers for IoT** April 2025 
DEEC Openlab, DEEC-IST, HackerSchool
- **Digital Electronics 101** 2023/2024/2025 
HackerSchool
- **Introduction to Digital Systems with the Vivado Suite workshop** Sep. 2024/2025 
DEEC
- **Arduino Sprint Workshop for first year students** Sep. 2025 
DEEC

ADDITIONAL INFORMATION

Languages: Portuguese (Native), English (Fluent), German (Basic), Russian (Basic)

PERSONAL INTERESTS AND PROJECTS

Interests: Reconfigurable Computing, Hacker Culture, Electronics Repair, Embedded Systems, Free and Open Source Software

- **SILk and SARACEn – HackLab Monitoring Co-Projects** 2024 – present
Themes: Microcontrollers, IoT, Sensors and Actuators, Printed Circuit Boards
 - Developed SILLk (Smart Integrated Lab Lock)
 - Supervised SILLK's PCB design
 - Developed SARACEn (Sensor Array for Reeavulation and Actuation in Controlled Environment)
 - Both SILLk and SARACEn will integrate with the HackerSchool website
- **GreenThumb – IoT Aeroponic System** 2023 – present
Themes: Microcontrollers, IoT, Sensors, Smart Urban Agriculture, Aeroponics
 - Designed an aeroponic plant growing system with sensor-based monitoring
 - Implemented embedded control and data acquisition
 - Designed enclosure (first prototype out of PET gallon)
- **jbr.pt – My self-hosted website and webserver** 2024 – present 
Themes: HTML, CSS, nginx
 - Designed my own custom and expandable website based on the hugo framework, using little JavaScript as requirement
 - Deployed from an old T410 Lenovo Thinkpad

REFERENCES

1. **Horácio Cláudio de Campos Neto**
Retired Professor with collaboration agreement, DEEC
Instituto Superior Técnico
Relationship: MSc and PhD Advisor
2. **Mário Pereira Véstias**
Coordinator Professor, DEETC
Instituto Superior de Engenharia de Lisboa
Relationship: MSc and PhD Advisor
3. **José João Teixeira de Sousa**
Associate Professor, DEEC
Instituto Superior Técnico
Relationship: PhD Advisor
4. **Rui Policarpo Duarte**
Adjunct Professor, DEETC
Instituto Superior de Engenharia de Lisboa
Relationship: Research Advisor
5. **Steven B. Leeb**
Emanuel E. Landsman Professor, EECS Dept.
Massachusetts Institute of Technology
Relationship: Senior collaborator (workshops)